Abnormal neuronal migration, deranged cerebral cortical organization, and diffuse white matter astrocytosis of human fetal brain: a major effect of...

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Detailed clinical and neuropathological studies have been made in two fullterm newborn human infants who were exposed to methylmercury in utero as a result of maternal ingestion of methylmercury-contaminated bread in early phases of pregnancy. High levels of mercury were detected in various regions of the brain at autopsy. Study of the brains revealed a disturbance in the development in both cases, consisting essentially of an incomplete or abnormal migration of neurons to the cerebellar and cerebral cortices, and deranged cortical organization of the cerebrum. There were numerous heterotopic neurons, both isolated and in groups, in the white matter of cerebrum and cerebellum and the laminar cortical pattern of the laminar cortical pattern of the cerebrum was disturbed in many regions as was shown by the irregular groupings and the deranged alignment of cortical. Prominent in the white matter of the cerebrum and the cerebellum was diffuse gemistocytic astrocytosis accompanied by an accumulation of mercury grains in their cytoplasm. These findings indicate a high degree of vulnerability of human fetal brain to maternal intoxication by methylmercury. A major effect appears to be related to faulty development and not to destructive focal neuronal damage as has been observed in mercury intoxicaiton in adults and children exposed postnatally.